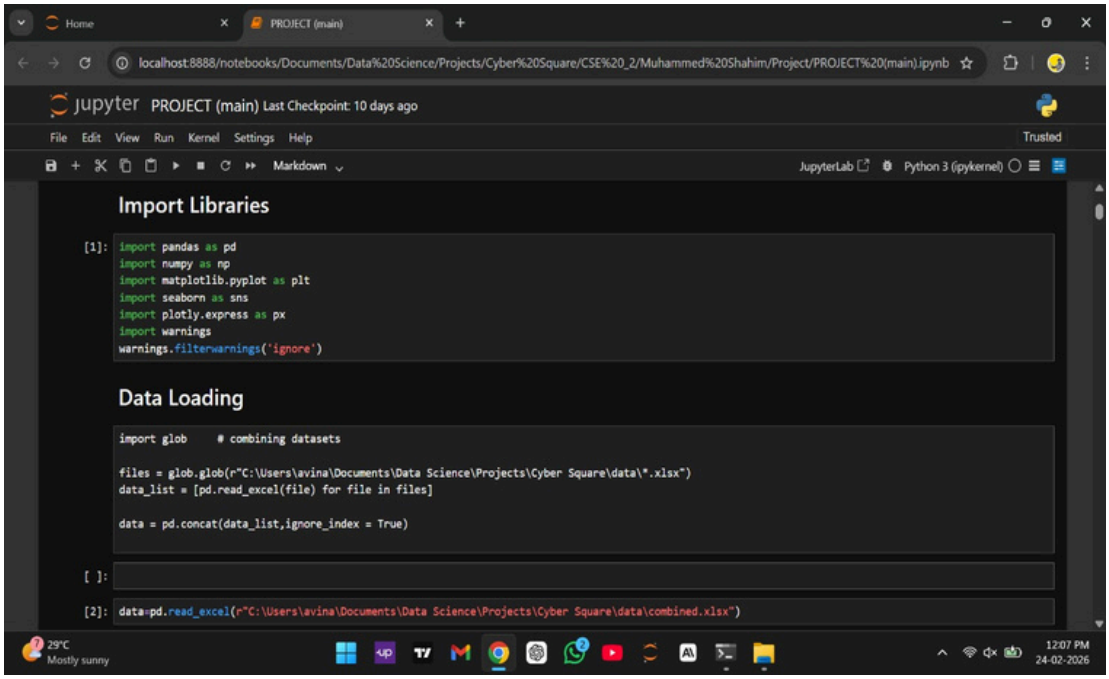
Project Data Cleaning Process Report

1) Import Required Libraries:

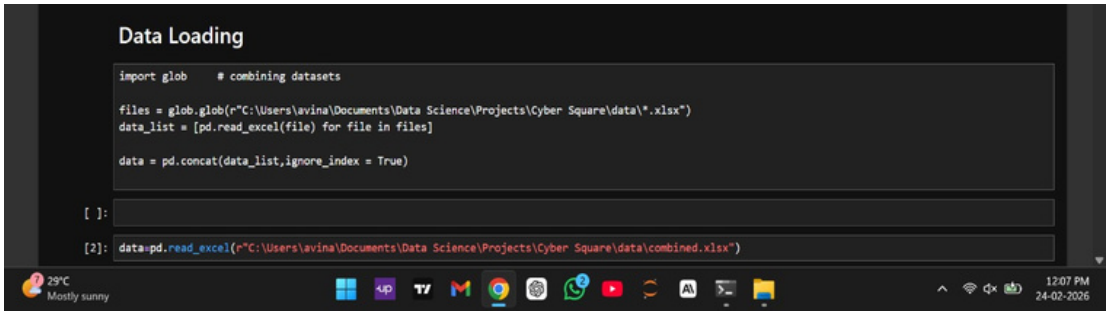
ThefollowingPythonlibrarieswereused for data cleaning:

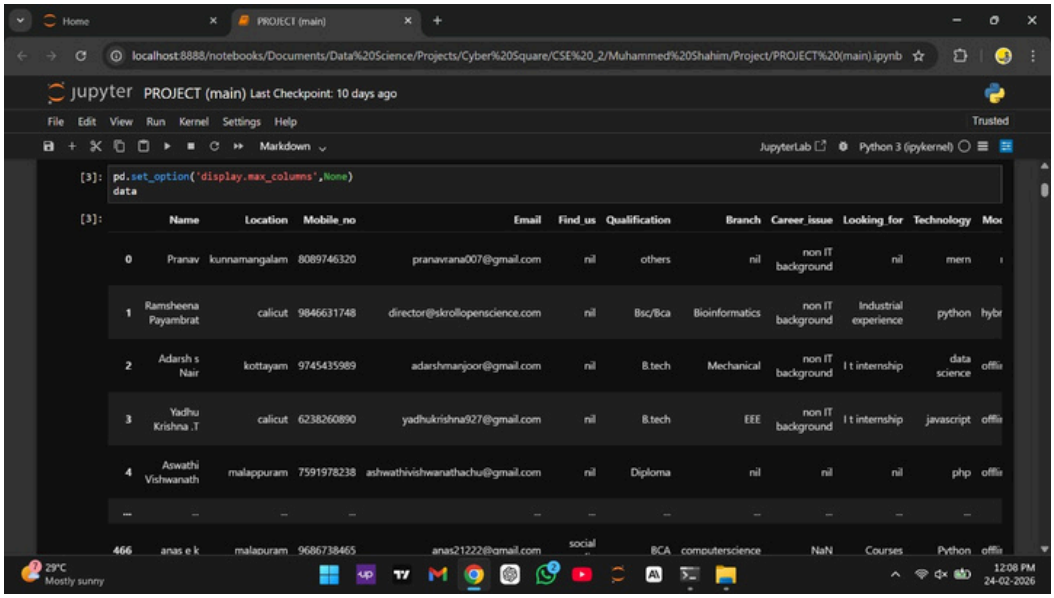
- Pandas :– Data manipulation and cleaning
- NumPy :– Numerical operations



2) Data Loading:

The data set was loaded from Excel files.





```
[3]: pd.set_option('display.max_columns',None)
     data
```

	Name	Location	Mobile_no	Email	Find_us	Qualification	Branch	Career_issue	Looking_for	Technology	Mo
0	Pranav	kunnamangalam	8089746320	pranavrana007@gmail.com	nil	others	nil	non IT background	nil	mern	
1	Ramsheena Payambrat	calicut	9846631748	director@skrollopenscience.com	nil	Bsc/Bca	Bioinformatics	non IT background	Industrial experience	python	hybr
2	Adarsh s Nair	kottayam	9745435989	adarshmanjoor@gmail.com	nil	B.tech	Mechanical	non IT background	I t internship	data science	offli
3	Yadhu Krishna .T	calicut	6238260890	yadhukrishna927@gmail.com	nil	B.tech	EEE	non IT background	I t internship	javascript	offli
4	Aswathi Vishwanath	malappuram	7591978238	ashwathivishwanathachu@gmail.com	nil	Diploma	nil	nil	nil	php	offli
...	...	...	...	...	...	...	...	...	...	...	
466	anas e k	malapuram	9686738465	anas21222@gmail.com	social	BCA	computerscience	NaN	Courses	Python	offli

- pd.set_option('display.max_columns', None) is used to show all columns of a dataset when displaying a DataFrame.
- It shows only a few columns with . . . in between.

3) Check basic dataset Informations

- data.info(): to identify data types

```
[4]: data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 471 entries, 0 to 470
Data columns (total 19 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Name              471 non-null    object
 1   Location          471 non-null    object
 2   Mobile_no         465 non-null    object
 3   Email             419 non-null    object
 4   Find_us           197 non-null    object
 5   Qualification     415 non-null    object
 6   Branch            378 non-null    object
 7   Career_issue      216 non-null    object
 8   Looking_for       354 non-null    object
 9   Technology        435 non-null    object
 10  Mode              273 non-null    object
 11  Time_Slot         260 non-null    object
 12  Date              449 non-null    object
 13  Attended_by       458 non-null    object
 14  No_persons        435 non-null    object
 15  Fee_Details       313 non-null    object
 16  Year_Of_pass_out  379 non-null    object
 17  Status            452 non-null    object
 18  Remarks           451 non-null    object
dtypes: object(19)
memory usage: 70.0+ KB
```

- data.columns: to display the names of all columns in a data

```
[5]: data.columns

[5]: Index(['Name', 'Location', 'Mobile_no', 'Email', 'Find_us', 'Qualification',
       'Branch', 'Career_issue', 'Looking_for', 'Technology', 'Mode',
       'Time_Slot', 'Date', 'Attended_by', 'No_persons', 'Fee_Details',
       'Year_Of_pass_out', 'Status', 'Remarks'],
      dtype='object')
```

4)Standardising column names:

- Removing extra spaces
- Converting letters to lowercase
- Making names consistent and readable (replace “ ” with “_”)

```
[6]: # standardising column names
     data.columns = ( data.columns
                     .str.strip()
                     .str.lower()
                     .str.replace(" ", "_"))
     data
```

[6]:

	name	location	mobile_no	email	find_us	qualification	branch	career_issue	looking_for	technology	mod
0	Pranav	kunnamangalam	8089746320	pranavrana007@gmail.com	nil	others	nil	non IT background	nil	mern	ni
	Ramsheena							non IT	Industrial		

5)Convert values to lowercase:

Converting values to lower case and also removes leading and trailing spaces.

```
[8]: # Converting values to lowercase and also removes leading and trailing spaces.

     data['name'] = data['name'].str.lower().str.strip()
     data['location'] = data['location'].str.lower().str.strip()
     data['find_us'] = data['find_us'].str.lower().str.strip()
     data['qualification'] = data['qualification'].str.lower().str.strip()
     data['branch'] = data['branch'].str.lower().str.strip()
     data['career_issue'] = data['career_issue'].str.lower().str.strip()
     data['looking_for'] = data['looking_for'].str.lower().str.strip()
     data['technology'] = data['technology'].str.lower().str.strip()
     data['time_slot'] = data['time_slot'].str.lower().str.strip()
     data['mode'] = data['mode'].str.lower().str.strip()
     data['attended_by'] = data['attended_by'].str.lower().str.strip()
     data['status'] = data['status'].str.lower().str.strip()
     data['remarks'] = data['remarks'].str.lower().str.strip()

[9]: data
```

6)Checking duplicates:

```
[10]: # checking duplicates

      data.duplicated().sum()

[10]: np.int64(0)
```

- Noduplicatespresentinthe data.

7) Handling Missing Values

1. Checking null values:

```
[11]: # checking null values present

      data.isna().sum()

[11]: name              0
      location          0
      mobile_no         6
      email            52
      find_us         274
      qualification    56
      branch           94
      career_issue    255
      looking_for     117
      technology       36
      mode            198
      time_slot       211
      date             22
      attended_by      13
```

- Identified the sum of each columns contain missing data.

```
[12]: null_count = data.isna().sum()
null_percent = (null_count / len(data)) * 100

null_table = pd.DataFrame({
    'Null_Count': null_count,
    'Null_Percentage (%)': null_percent.round(2)
})

null_table
```

[12]:

	Null_Count	Null_Percentage (%)
name	0	0.00
location	0	0.00
mobile_no	6	1.27
email	52	11.04
find_us	274	58.17
qualification	56	11.89
branch	94	19.96

- display the number and percentage of missing values in each column of the dataset in a clear table format.

```
[13]: # Standardising multiple missing-value representations into a single NaN format.

data = data.replace(['nil', 'nan','np','NaN', 'NULL', 'null','', ' '], np.nan)

[ ]:
```

- Standardising multiple missing-value representation into a single NaN format.

Clean Location Column:

Identify all unique values in the location column of a dataset.

```
[14]: data['location'].unique()

[14]: array(['kunnamangalam', 'calicut', 'kottayam', 'malappuram', 'kondotty',
       'chaliyam', 'farook college', 'bc road beypore', 'farook',
       'beypore', 'methottuthazha', 'pantheerankavu', 'manjeri',
       'vadaksra', 'payyanakkal', 'wayanad', 'kuttikatur', 'payyoli',
       'medical college', nan, 'palazhi', 'kovoor', 'thikkody',
       'vazhayoor', 'kannur', 'areekode', 'tirurangadi', 'koyilandy',
       'ramanatukkara', 'edavannapara', 'marayur', 'mukkam', 'balussery',
       'feroke', 'parambilbazar', 'calcu', 'ulliyeri', 'thalaserry',
       'kommeri', 'thamssery', 'perambra', 'kallayi', 'quilandi',
       'vadakara', 'thrissur', 'poovangad', 'kallai', 'kuttikkaattor',
       'cheruvannur', 'tirur', 'ramanattukara', 'malapuram', 'tanur',
       'karaparambu', 'trivandrum', 'kozhikode', 'panthirankaav',
       'vengeri', 'naduvannur', 'koduvally', 'karaparamba', 'valanchery',
       'kuttippuram', 'nit', 'kuttyadi', 'thondayad', 'puthiyangadi',
       'kadalundi', 'vallikkunnu', 'areepara', 'thiruvannur', 'mahe',
       'parappanangadi', 'olavanna', 'chevayoor', 'puthuppamba',
       'kadavathur', 'vellimadukkunnu', 'palakkad', 'kannookkara',
       'kasargod', 'ponnani', 'kankot', 'pandikkad', 'puthurmadam',
       'pokkunnu', 'nadapuram', 'artholi', 'mundayad', 'palakad',
       'kotholi', 'mukkonu', 'chathamangalam', 'pullavoor', 'ayikarapadi',
```

```
[15]: data['location'] = data['location'].replace(
    {'artholi' : 'atholi',
     'arikode':'areekode',
     'bc road beypore' : 'beypore',
     'calcu' : 'calicut',
    'chevayoor' : 'chevayur',
    'edavannapara' : 'edavannappara',
    'feroke' : 'feroke',
    'farook' : 'feroke',
    'kallayi' : 'kallai',
    'karaparambu' : 'karaparamba',
    'katholi' : 'atholi',
    'kotholi' : 'atholi',
    'koyilandi' : 'koyilandy',
     'kondotty':'kondoty',
    'kozhikode' : 'calicut',
    'kozhikkode' : 'calicut',
    'kuttikatur' : 'kuttikkaattor',
    'maankavu' : 'mankavu',
    'malapuram' : 'malappuram',
    'mlappuram' : 'malappuram',
    'mankave' : 'mankavu',
     'marthottam':'mathottam',
    'nadakavu' : 'nadakkavu',
    'thirur' : 'tirur',
    'palakad' : 'palakkad',
```

```
    'palakad' : 'palakkad',
    'pantheerankave' : 'pantheeramkavu',
    'pantheerankavu' : 'pantheeramkavu',
    'panthirankaav' : 'pantheeramkavu',
    'parambil' : 'parambilbazar',
    'quiland' : 'koyilandy',
    'quilandy' : 'koyilandy',
    'ramanattukkara' : 'ramanattukara',
    'thamssery' : 'thamarassery',
    'thamaressery' : 'thamarassery',
    'thammarassery' : 'thamarassery',
    'thenkipalam' : 'thenjipalam',
    'vadaksra' : 'vadakara',
    'vazhayoor' : 'vazhayur',
    'plarkunnu' : 'palakunnu',
     'vellimadkunnu':'vellimadukkunnu',
     'vengra':'vengara',
    'chalcode' : 'chalakudy',
    'nil':'nit',
    'thirur':'tirur',
    'pokallai':'kallai',
    'cheruvayoor':'chevayur',
    'madantavida':'mananthavadi',
     'ernhipalam':'eranhipalam',
     'kalad':'kaladi'
    })
```

- Checked unique values in the location feature and cleaned by replacing with real ones.

```
[16]: data['location'] = data['location'].fillna(data['location'].mode()[0])

[17]: data['location'].isna().sum()

[17]: np.int64(0)
```

- Null values in the column is filled by mode (calicut).

Clean ‘email’ and ‘mobile_no’ column:

- These features are non-critical, replaced missing values with meaningful placeholder.

```
[18]: data['email'] = data['email'].fillna('unknown')
      data['mobile_no'] = data['mobile_no'].fillna('unknown')
```

Clean ‘find_us’ column:

```
[19]: data['find_us'].unique()

[19]: array([nan, 'google', 'social media', 'reference', 'others', 'i', 'other'],
            dtype=object)
```

- Checked unique values in the ‘find_us’ column.

```
[20]: data['find_us'] = data['find_us'].replace({
          'refernce':'reference',
          'i':'social media',
          'instagram':'social media',
          'others':'other'
```

- Corrected spelling mistake to maintain category accuracy
- ‘i’ and ‘instagram’ are also social media platforms, that are replace with ‘social media’
- Since missing value exceed 20%, null values were remained unchanged for model reliability.

Clean ‘qualification’ column:

```
[22]: data['qualification']=data['qualification'].replace(
          {'b tech':'b.tech/m.tech',
          'others':'other',
          'b.tec':'b.tech/m.tech',
           'b.tech':'b.tech/m.tech',
           'bba':'ug',
           'm.tec':'b.tech/m.tech',
           'ba':'ug',
           'ma':'ug',
           'b.com':'ug',
          'commerce':'ug',
          'plus 2':'plus two',
           'plus two':'plus two',
          'bcom':'ug',
          'bvoc':'ug',
          'bsc':'ug',
          'non it':'other',
          'msc':'pg',
           'mca':'pg',
           'bca':'pg',
          'pg':'pg',
           'B.tech':'b.tech/m.tech',
           'mba':'pg',
           'm.tech':'b.tech/m.tech',
          'ug':'ug',
          'bba/bcom':'ug',
```

```
           'bba':'ug',
           'm.tec':'b.tech/m.tech',
           'ba':'ug',
           'ma':'ug',
           'b.com':'ug',
          'commerce':'ug',
          'plus 2':'plus two',
           'plus two':'plus two',
          'bcom':'ug',
          'bvoc':'ug',
          'bsc':'ug',
          'non it':'other',
          'msc':'pg',
           'mca':'pg',
           'bca':'pg',
          'pg':'pg',
           'B.tech':'b.tech/m.tech',
           'mba':'pg',
           'm.tech':'b.tech/m.tech',
          'ug':'ug',
          'bba/bcom':'ug',
           'mba/mcom':'pg',
          'msc/mca':'pg',
          'bsc/bca':'ug',
         'b.tech/m.tech.':'b.tech/m.tech'})
```

- Values in the qualification column is grouped into 7 categories.

```
[23]: data['qualification'].unique()

[23]: array(['other', 'ug', 'b.tech/m.tech', 'diploma', nan, 'plus two', 'pg',
       'sslc'], dtype=object)

[24]: data['qualification'].isna().sum()

[24]: np.int64(73)

[25]: data['qualification'] = data['qualification'].fillna(method='bfill')

[26]: data['qualification'].isna().sum()

[26]: np.int64(0)
```

- Null values in the column are filled by bfill method.

Clean ‘branch’ column:

```
[27]: data['branch'].unique()

[27]: array([nan, 'bioinformatics', 'mechanical', 'eee', 'cs',
       'business administration', 'ba sociology', 'b.com', 'i t', 'ece',
       'finance', 'graphic web', 'bca', 'cse', 'honars n mathematics',
       'science', 'zoology', 'ai', 'economics', 'humanities', 'english',
       'civil', 'applicaton', 'electronics', 'physics', 'it',
```

- Checked unique value in the ‘branch’ feature.

```
data['branch']=data['branch'].replace(
    {'aei':'ai',
   'artificial intelligence':'ai',
    'automobile':'automobile engineering',
    'bca':'computer application',
    'cs':'computer science',
    'applicaton':'computer application',
    'ba english':'english',
    'computerscience':'computer science',
    'mca':'computer application',
    'psycology':'psychology',
    'zology':'zoology',
    'ec':'electronics',
    'e&c':'electronics',
    'ct':'computer science',
    'bsc computer science':'computer science',
    'm.com':'commerce',
    'ca':'computer application',
    'automobile':'automobile engineering',
    'mba':'business administration',
    'bsc electronics':'electronics',
    'bsc psychology':'psychology',
    'mech':'mechanical engineering',
    'b .com ca':'computer application',
    'thermal engineering':'mechanical engineering',
    'electronics and communication':'electronics',
```

```
    'thermal engineering':'mechanical engineering',
    'electronics and communication':'electronics',
    'plustwo':'science',
    'plus 2':'science',
    'plus two':'science',
    'bvoc':'it',
    'b.voc':'it',
    'mechanical':'mechanical engineering',
    'electrical':'electrical engineering',
    'civil':'civil engineering',
    'bsc electronics':'electronics',
    'ce':'computer engineering',
    'ece':'electronics',
    'bcom':'commerce',
    'eee':'electronics',
    'civil engineeering':'civil engineering',
    'physiscs':'physics',
    'finance':'commerce',
    'bsc economics and maths':'economics',
    'b com':'commerce',
    'computer science and engineering':'computer engineering',
    'computer hardware and networking':'computer application',
    'i t':'it',
    'honars n mathematics':'mathematics',
    'computer hardware':'computer application',
    'ba sociology':'sociology',
```

```
    'bsc electronics':'electronics',
    'ce':'computer engineering',
    'ece':'electronics',
    'bcom':'commerce',
    'eee':'electronics',
    'civil engineeering':'civil engineering',
    'physiscs':'physics',
    'finance':'commerce',
    'bsc economics and maths':'economics',
    'b com':'commerce',
    'computer science and engineering':'computer engineering',
    'computer hardware and networking':'computer application',
    'i t':'it',
    'honars n mathematics':'mathematics',
    'computer hardware':'computer application',
    'ba sociology':'sociology',
    'cse':'computer engineering',
    'video editing':'videography',
    'computer enginering':'computer engineering',
    'b.com':'commerce',
    'bba':'business administration',
    'business':'business administration'})

data['branch'] = data['branch'].replace(['msc','bsc'], np.nan)
```

- Corrected spellingmistakes to maintain categoryaccuracy.
- Since missing values exceed 20%, null values were kept unchanged for model reliability.

Clean ‘career_issue’ column:

```
[30]: data['career_issue'].unique()

[30]: array(['non it background', nan, 'career gap', 'backpaper', 'backpapers',
       'non it', 'careergap', 'non-it'], dtype=object)
```

- Checked unique value in the ‘career_issue’ column.

```
[31]: data['career_issue']=data['career_issue'].replace(
          {'backpapers':'backpapers',
           'backpapper':'backpapers',
           'backpappers':'backpapers',
           'non-it':'non it background',
           'non it':'non it background',
           'Backpaper':'backpapers',
           'carrer gap':'career gap',
           'careergap':'career gap',
           'backpaper':'backpapers'})

[32]: data['career_issue'].unique()

[32]: array(['non it background', nan, 'career gap', 'backpapers'], dtype=object)
```

- Values in the ‘career_issue’ column is grouped into 3 categories.
- Null values in the column kept unchanged (>20%)

Clean ‘looking_for’ column:

```
[33]: data['looking_for'].unique()

[33]: array([nan, 'industrial experience', 'i t internship',
             'short term course', 'it internship', 'it work experience',
             'it courses', 'others', 'it crash course', 'internship',
             'job orineted course', 'job oriented course',
             'it  work experience', 'it course', 'iv', 'work exprience',
```

- Chcked unique value in the ‘looking_for’ column.

```
[34]: data['looking_for']=data['looking_for'].replace(
          {'i t internship':'it internship',
           'internship':'it internship',
           'intership':'it internship',
           'job orineted course':'it work experience',
           'job oriented course':'it work experience',
           'work experience':'it work experience',
           'it crash course':'it course',
          'industrial experience':'it work experience',
          'it courses':'it course',
          'short term course':'it course',
          'courses':'it course',
          'iv':'it course',
          'it work experince':'it work experience',
          'others':'other',
          'it  work experience':'it work experience',
          'work exprience':'it work experience'})

[35]: data['looking_for'].unique()

[35]: array([nan, 'it work experience', 'it internship', 'it course', 'other'],
            dtype=object)
```

- Values in the ‘looking_for’ column is grouped into mainly 3 categories.
- Null values in the column kept unchanged (>20%)

Clean ‘technology’ column:

```
[36]: data['technology'].unique()

[36]: array(['mern', 'python', 'data science', 'javascript', 'php', 'devops',
             'other', 'business analytics', nan, 'flutter', 'react', 'ai',
             'others', 'web designing', 'angular', 'artificial intelligence',
             'java', 'data analysis', 'java script'], dtype=object)
```

- Chcked unique value in the ‘technology’ column.

```
[37]: data['technology']=data['technology'].replace({
          'data analysis':'data analytics',
          'artificial intelligence':'ai',
          'bussines analystics':'business analytics',
          'business analystic':'business analytics',
          'datascience':'data science',
          'ai':'ai/ml',
          'javascript':'java script',
          'java scirpt':'java script',
          'artificial intelligence': 'ai/ml',
          'web designing':'ui/ux',
          'others':'other'})

[38]: data['technology'].unique()

[38]: array(['mern', 'python', 'data science', 'java script', 'php', 'devops',
             'other', 'business analytics', nan, 'flutter', 'react', 'ai/ml',
             'ui/ux', 'angular', 'java', 'data analytics'], dtype=object)
```

- Corrected spelling mistake to maintain category accuracy.

```
[39]: data['technology'].mode()[0]

[39]: 'python'

[40]: data['technology'] = data['technology'].fillna(data['technology'].mode()[0])
```

- Null values were filled by mode (python)

Clean ‘mode’ column:

```
[41]: data['mode'].unique()
[41]: array([nan, 'hybrid', 'offline', 'online', '0f'], dtype=object)
[42]: data['mode'] = data['mode'].replace({'0f' : 'offline'})
[43]: data['mode'].unique()
[43]: array([nan, 'hybrid', 'offline', 'online'], dtype=object)
```

- Values in ‘mode’ column is grouped into 3 categories.
- Replace the ‘0f’ with ‘offline’.
- Nullvalues inthe column kept unchanged (>20%)

Clean ‘time_slot’ column:

```
[44]: data['time_slot'].unique()
[44]: array([nan, 'morning', 'afternoon', 'nil;', 'weekend', 'evening'],
          dtype=object)
[45]: data['time_slot'] = data['time_slot'].replace({
          'morninng':'morning',
          'nil;':'afternoon'
      })
[46]: data['time_slot'].unique()
[46]: array([nan, 'morning', 'afternoon', 'weekend', 'evening'], dtype=object)
```

- Values in the ‘time_slot’ column is grouped into 4 categories.
- Corrected the spelling mistake and replace ‘nil’ with ‘afternoon’.
- Nullvalues in thecolumn kept unchanged (>20%)

Clean ‘date’ column:

```
[47]: data['date'] = data['date'].bfill()
[48]: data['date'].isna().sum()
[48]: np.int64(0)
```

- Null values in the date column is filled by using bfill method.
- Corrected dateformat (eg:- ’12-01-2005’:’12-01-2025’)

Adding new column (year):

```
[50]: data['date'] = pd.to_datetime(data['date'], errors='coerce')
      data['year'] = data['date'].dt.year
      data['month'] = data['date'].dt.month_name()
[51]: data['year'].unique()
[51]: array([2024, 2023, 2025, 2021, 2022, 2002, 2005], dtype=int32)
[52]: data['year'] = data['year'].replace({2002:2022,2005:2025})
[53]: data['year'].unique()
[53]: array([2024, 2023, 2025, 2021, 2022], dtype=int32)
```

- new column (year) was added to dataframe by extracting from date.

Clean ‘year_of_pass_out’ column:

```
[55]: data['year_of_pass_out'].isna().sum()

[55]: np.int64(0)

[56]: data['year_of_pass_out'] = data['year_of_pass_out'].replace({
        2026. : 2016.,
         2027. : 2017.
      })

[57]: data['year_of_pass_out'].unique()

[57]: array([2021., 2020., 2022., 2018., 2024., 2023., 2016., 2025., 2019.,
             2015., 2017., 2013., 2012., 2014., 2010., 2004., 2011., 2008.])
```

- null values in the ‘year_of_pass_out’ column is filled with ffill.
- Errors in the data replaced with real ones.

Clean ‘attented_by’ column:

```
[59]: data['attended_by'].unique()

[59]: array(['baiju', 'rifana', 'praveena', 'shivaprasad', nan, 'anusree',
             'sivaprasad', 'anupama', 'shafeeque', 'deepak', 'monish', 'rifna',
             'athulya', 'remin', 'shiju', 'lekha', 'faisal', 'lekha r',
             'shiva prasad', 'legha'], dtype=object)

[60]: data['attended_by']=data['attended_by'].replace({
          'sivaprasad':'shivaprasad',
          'shiva prasad':'shivaprasad',
          'sivaprasd':'shivaprasad',
          'biju':'baiju',
          'lekha r':'lekha',
          'legha':'lekha'
      })

[61]: data['attended_by'].mode()[0]

[61]: 'baiju'

[62]: data['attended_by'] = data['attended_by'].fillna(data['attended_by'].mode()[0])
```

- Checked unique values in the ‘attented_by’ column.
- Nullvaluesinthe data is filled by using mode (‘baiju’)

Clean ‘status’ column:

```
[63]: data['status'].unique()

[63]: array(['yes', 'no', nan, 'cancelled', 'joined', 'not joined'],
            dtype=object)

[64]: data['status'] = data['status'].replace({
          'not joined':'no',
          'joined':'yes',
          'cancelled':'no',
          'n':'no'
      })
```

- Checked unique values in status and grouped to 2 categories, ‘yes’ for joined and ‘no’ for not joined.
- Null values kept unchanged for better reliability.

Clean ‘remarks’ column:

```
[65]: data['remarks']=data['remarks'].replace({
        'went to abudhabi':'went to abroad',
        'going abroad':'went to abroad',
        'No update':'not responding',
        'jined':'joined',
        'forwarded to arabic':'not responding',
        'come for enquiry':'came for enquiry',
        'just enquired':'came for enquiry',
        'canceled':'not interested',
        'cancelled':'not interested',
        'not looking it':'not interested',
        'not intrested':'not interested',
        'number busy':'not responding',
      'joined other institution':'joined another institution',
        'found other':'joined another institution',
    'enquiry for job':'came for enquiry',
    'disonnected':'not responding',
    'enquiry for friend':'came for enquiry',
    'joined at kondotty sql':'joined another institution',
    'rush shedule':'rush schedule',
    'not contacts':'not responding',
    'not picking':'not responding',
    'call not picking':'not responding',
    'no':'not responding',
    'np': 'not joined',
    'pg':'going for higher studies',
```

```
Markdown
        'canceled':'not interested',
        'cancelled':'not interested',
        'not looking it':'not interested',
        'not intrested':'not interested',
        'number busy':'not responding',
      'joined other institution':'joined another institution',
        'found other':'joined another institution',
    'enquiry for job':'came for enquiry',
    'disonnected':'not responding',
    'enquiry for friend':'came for enquiry',
    'joined at kondotty sql':'joined another institution',
    'rush shedule':'rush schedule',
    'not contacts':'not responding',
    'not picking':'not responding',
    'call not picking':'not responding',
    'no':'not responding',
    'np': 'not joined',
    'pg':'going for higher studies',
    'sample enquiry':'came for enquiry',
    'no contacts':'not responding',
    'closed lost':'not joined',
    'suspected softronix':'suspected',
    'no update':'not responding',
    'not joined':'unknown',
    'low fee':'joined'})
```

```
[66]: data['remarks'].unique()

[66]: array(['joined', 'suspected', nan, 'not joined', 'not interested',
             'joined another institution', 'not applicable', 'rush schedule',
             'not responding', 'having backpapers', 'looking for ui/ux',
             'no certificate needed', 'faculty shortage', 'digital marketing',
             'waiting for interview result', 'going for higher studies',
             'came for enquiry', 'went to abroad', 'unknown', 'fake',
             'high fee', 'will update'], dtype=object)
```

- Replace the remarks column unique values in a proper way.
- Nullvalues inthecolumnkept unchanged (>20%)

Clean ‘no_persons’ column:

```
[71]: data['no_persons']=data['no_persons'].fillna(data['no_persons'].mode()[0])
```

- Null values in the column is filled by using mode (1)

```
[72]: data.isna().sum()

[72]: name                0
      location            0
      mobile_no           0
      email               0
      find_us           335
      qualification       0
      branch            135
      career_issue      313
      looking_for       140
      technology          0
      mode              225
      time_slot         248
      date                0
      attended_by         0
      no_persons          0
      fee_details       194
      year_of_pass_out    0
      status             19
```

Visualisation Part

SECTION 1: Geographic & Location Analysis

1.1 Top 10 Student Locations

The barchartbelow showsthe top 10locations by studentenquiry volume. The top location leads with approximately 125 enquiries, followed by the second (~110) and third (~103). The remaining locations each contribute 30–45 enquiries.

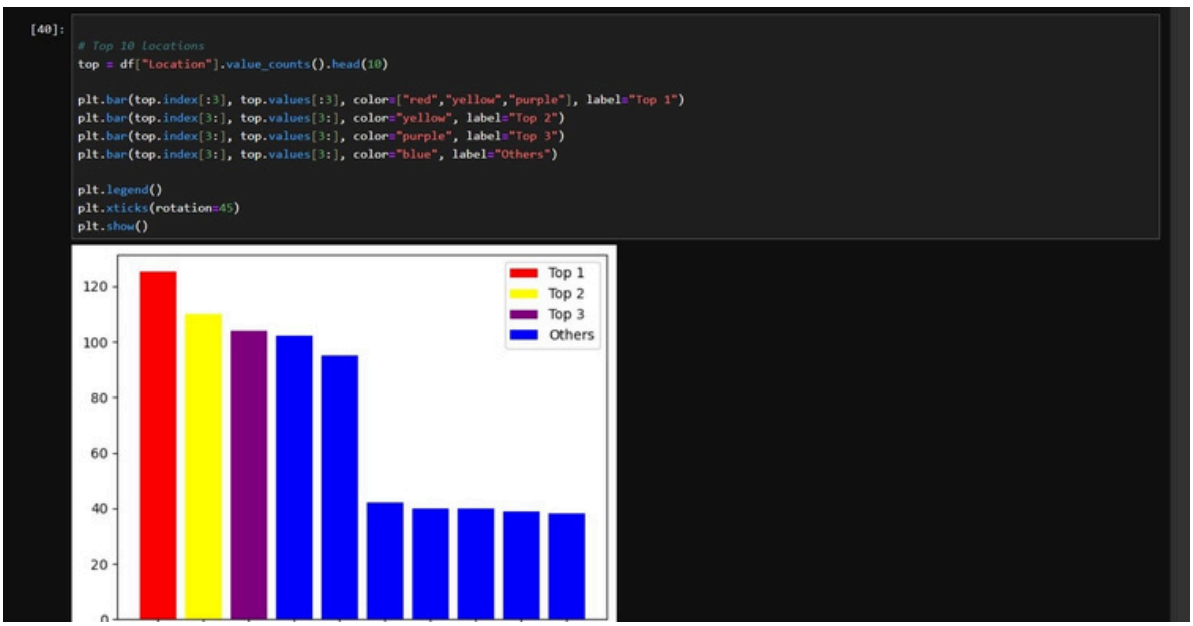


Chart 1: Top 10 Student Locations (Red = Top 1, Yellow = Top 2, Purple = Top 3, Blue = Others)

Key Findings:

- The top 3 locations collectively dominate enquiry volume — a clear geographic concentration.
- A sharp drop-off exists between the top 3 and the remaining 7 locations.
- The institute should intensify marketing and consider satellite centres in the top 3
- locations.

  Locations outside the top 3 each contribute 30–45 students, indicating broad but diluted geographic reach.

SECTION 2: Qualification & Academic Background

2.1 Qualification-wise Student Distribution (Pie Chart)

Thedonut chart showsthe proportionofstudents byqualificationlevel. UG students form the dominant group, comprising nearly two-thirds of all enquiries.

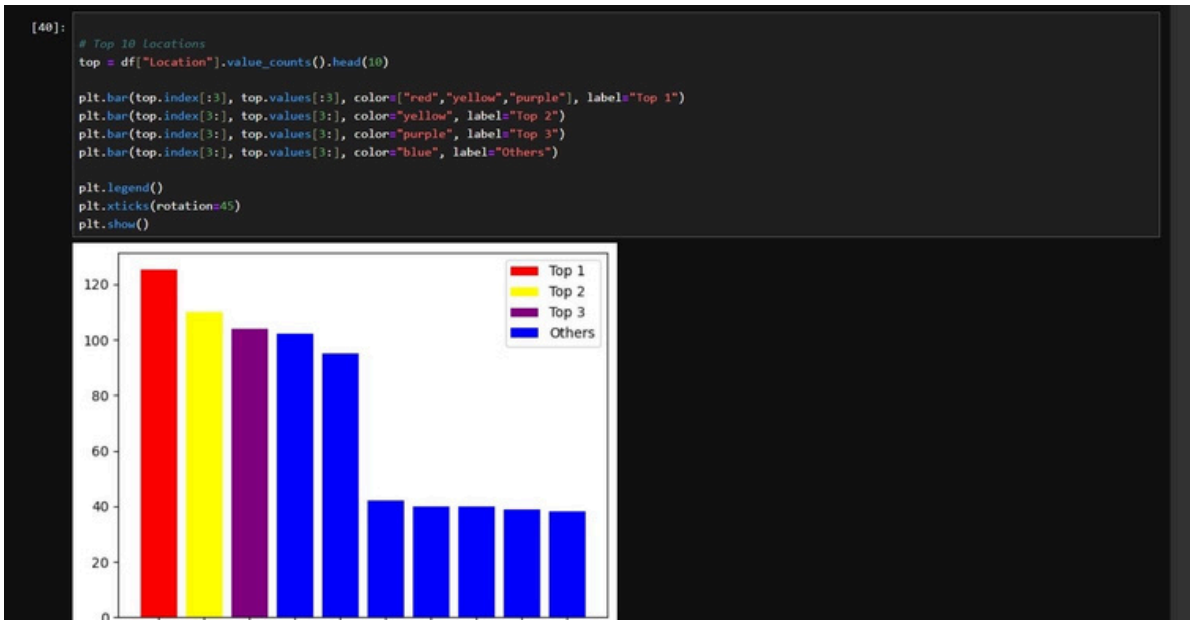


Chart 2: Qualification-wise Student Distribution — UG: 64.1%, PG: 20.1%, Plus 2: 4.64%, Diploma: 3.77%, Others: 3.77%, SSLC: 3.55%

Key Findings:

- UG students account for 64.1% of all enquiries — the single largest group by far.
- PG students are the second largest segment at 20.1%.
- Plus 2, Diploma, Others, and SSLC combined make up only ~15.7%, indicating limited outreach to pre-degree students.

2.2 Qualification-wise Student Count by Status

Thisgrouped barchartcomparesthenumberofJoinedvsNot Joined students across each qualification category.

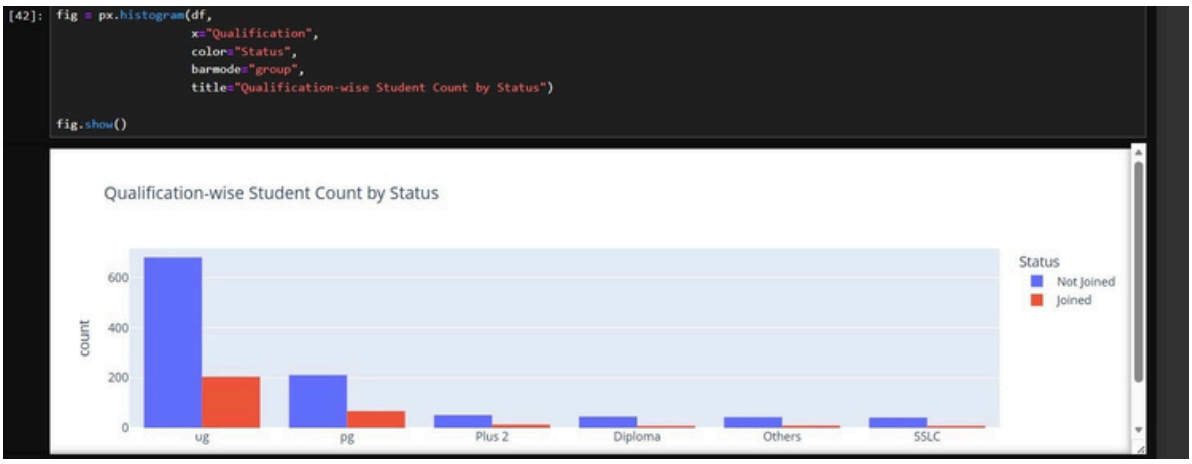


Chart 3: Qualification-wise Student Count by Status (Blue = Not Joined, Red = Joined)

Key Findings:

- UG students have the highest absolute joined count (~200) but also the highest not-joined count (~660), a ratio of roughly 1:3.
- PG students show approximately 55 joined vs 210 not joined — a similar 1:4 ratio.
- Plus 2, Diploma, Others, and SSLC categories show very few conversions, suggesting these groups need targeted outreach.

SECTION 3: Branch-wise Analysis

3.1 Branch Type Count

Thisbarchartshowsthetotal number of students across six academic branches: commerce, electronics, other, computer science, civil, and mechanical.

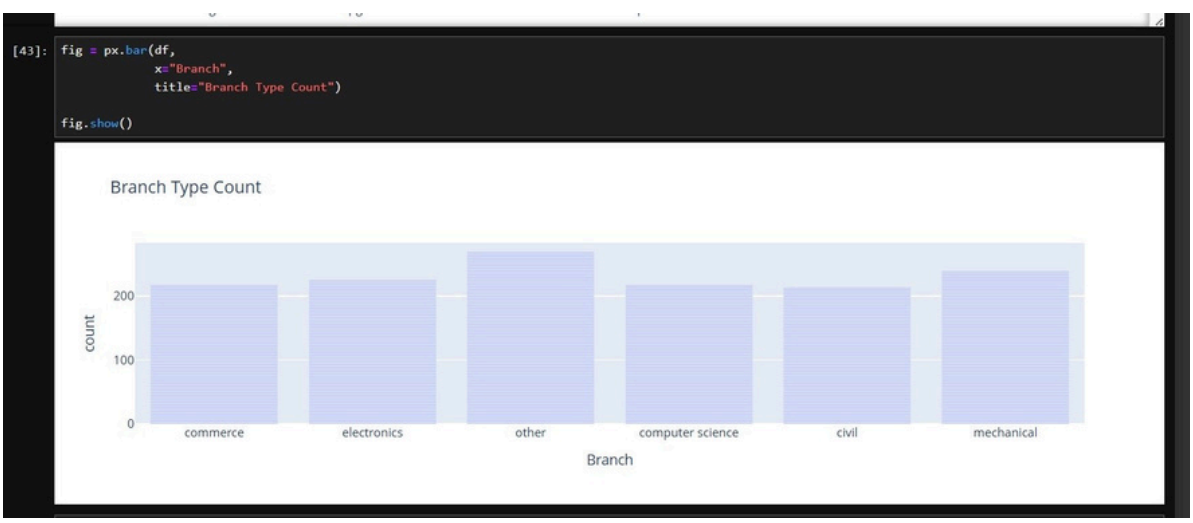


Chart 4: Branch Type Count — fairly uniform distribution across all six branches

Key Findings:

- The 'Other' branch leads with approximately 270 students, followed by electronics (~240) and mechanical (~240).
- Commerce, computer science, and civil each contribute around 215–225 students.
- The near-equal distribution indicates the institute attracts students from diverse academic backgrounds.

3.2 Branch-wise Status Distribution

ThisgroupedbarchartbreaksdownJoinedvsNot Joined students within each branch.

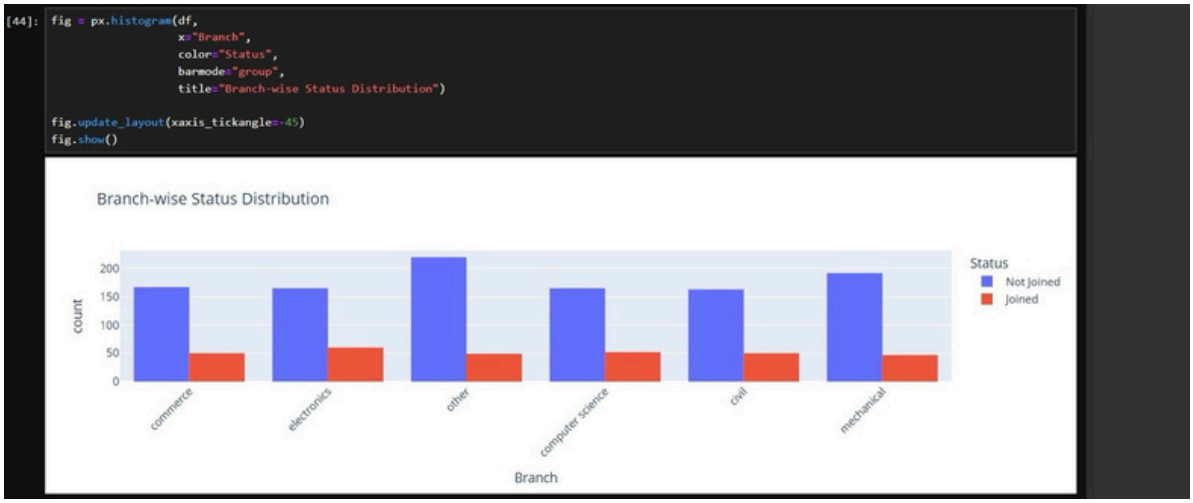


Chart 5: Branch-wise Status Distribution (Blue = Not Joined, Red = Joined)

Key Findings:

- Acrossall branches, Not Joined students significantly outnumber Joined students — consistently around 3:1.
- Electronics has a slightly higher Joined count (~60) compared to other branches (~45–55).
- The 'Other' branch has the highest Not Joined count (~210), reflecting its larger total volume.
- No branch shows a significantly better conversion rate, pointing to systemic barriers rather than branch-specific issues.

SECTION 4: Year of Pass-out Analysis

4.1 Year of Passout Count (Chronological)

ThisbarchartshowsthestudentcountsortedchronologicallyBy graduation year (2015–2025).

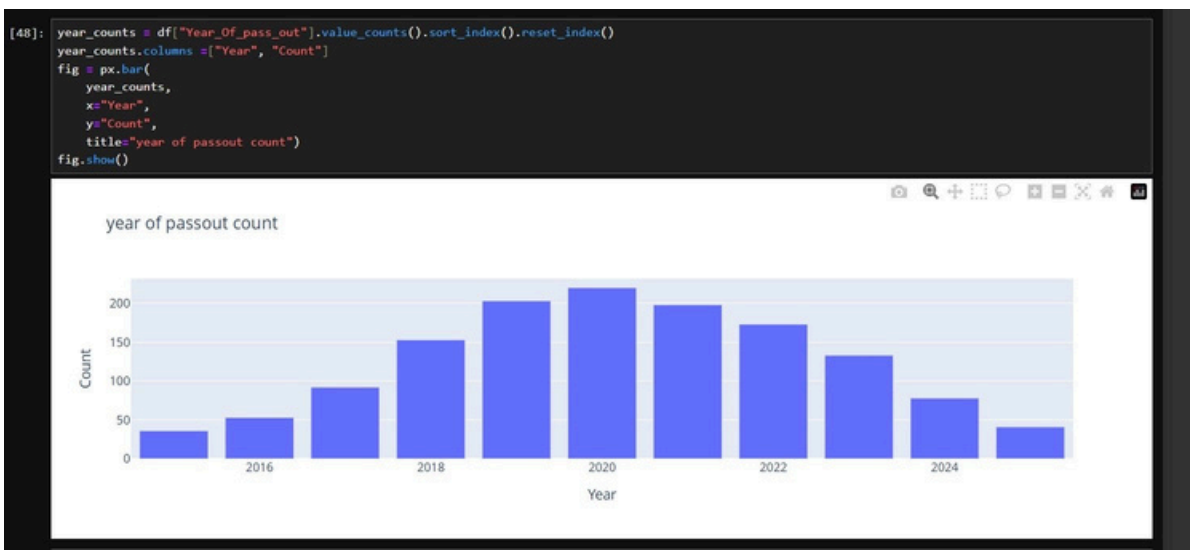


Chart 6: Year of Passout Count — sorted chronologically from 2015 to 2025

Key Findings:

- Enquiry volume grew steadily from 2015 (~35) through a peak in 2020 (~215), then declined.
- The 2020 peak likely reflects pandemic-driven career uncertainty prompting re-skilling interest.
- Recent graduates (2023–2025) show lower counts, possibly due to delayed awareness or outreach gaps.

4.2 Year of Pass-out Distribution (Count-sorted)

Thisalternativeviewsortsyearsbyfrequency,confirming2020 graduates as the most represented cohort.

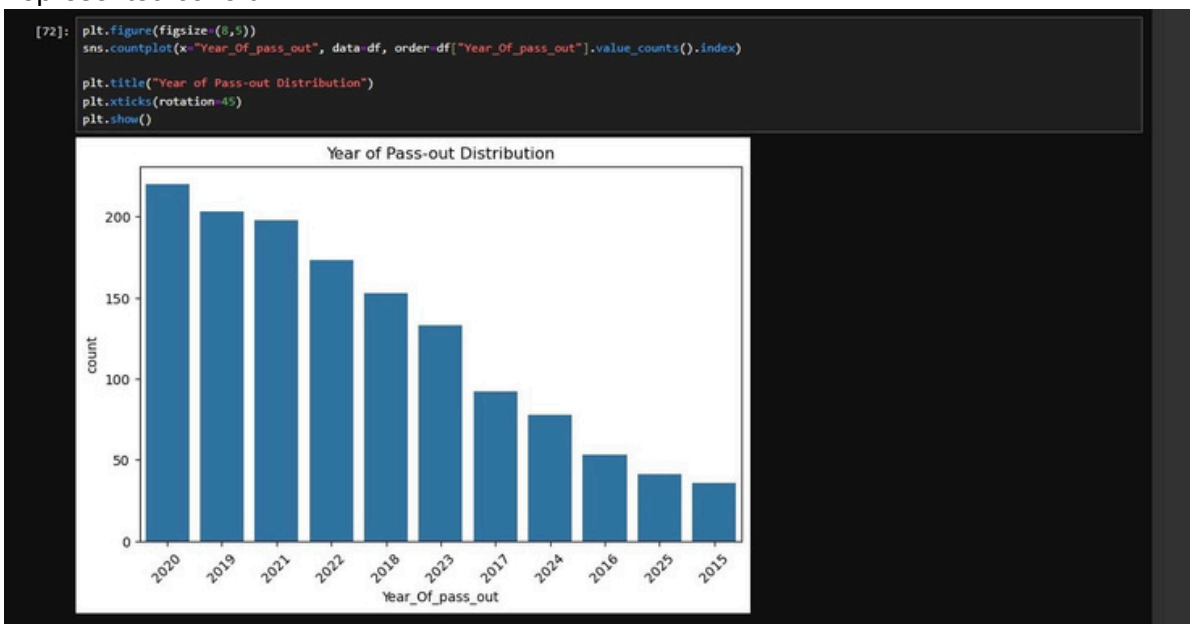


Chart 7: Year of Pass-out Distribution — sorted by count (2020 highest at ~220, 2015 lowest at ~37)

Key Findings:

- 2020 (220), 2019 (204), and 2021 (198) are the top three cohorts. 2015 and 2025 graduates are the smallest groups, at the extremes of the dataset range.
- The institute should develop alumni-style re-engagement campaigns targeting the 2017–2019 cohort who may now be seeking career transitions.

SECTION 5: Career Issues & Student Goals

5.1 Career Issues Distribution (Bar Chart)

Thisplotlybarchartshowsthefourcareerissuecategories and their respective student counts.

Chart 8: Career Issues Distribution — No Issues leads at ~700, followed by Career Gap (~290), Non-IT Background (~205), Back Papers (~130)

Key Findings:

- The majority of students (~700) report 'No Issues', indicating a relatively clean academic/professional profile.
- 'Career Gap' is the second most common issue (~290), reflecting students re-entering the job market.

5.2 Career Issues Distribution (Horizontal Bar)

Thisseabornhorizontalbarchartprovidesanalternativeview of the same data, ordered by count.

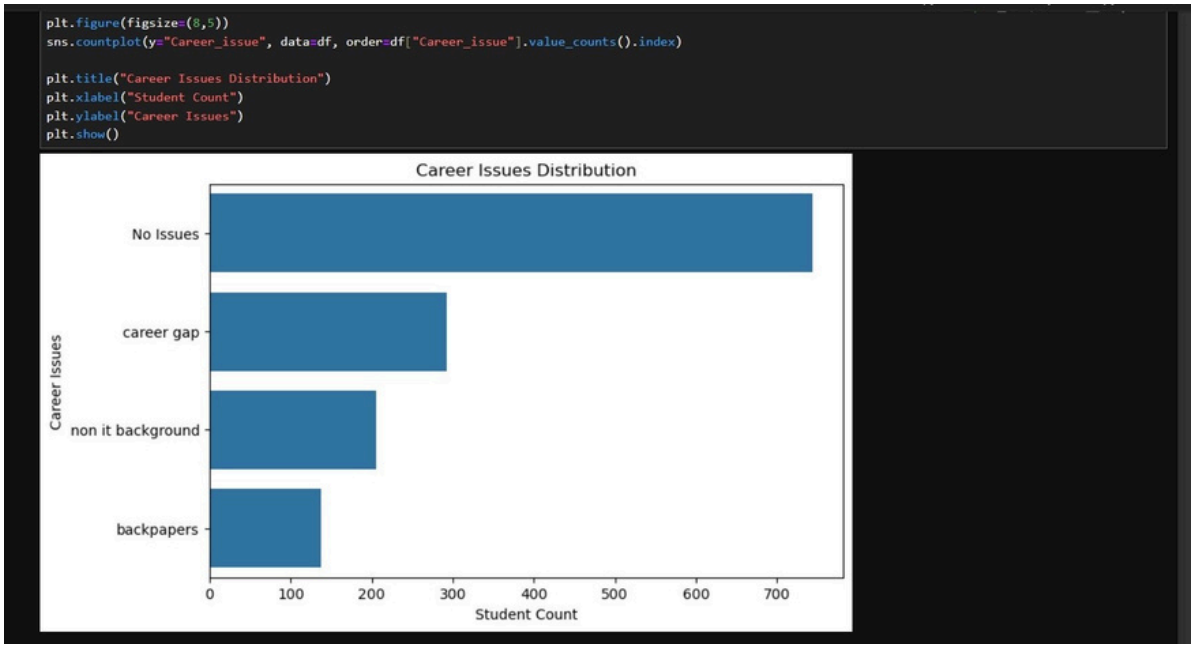


Chart 9: Career Issues Distribution (Horizontal) — confirms No Issues dominance at ~750 student count

Key Findings:

- No issues dominate (~750 students)
- Career gap is the top issue (~300)
- Non-IT background is a key challenge (~200)
- Backlogs affect fewer students (~150)
- Data is highly imbalanced

5.3 Career Issue vs Enrollment Status

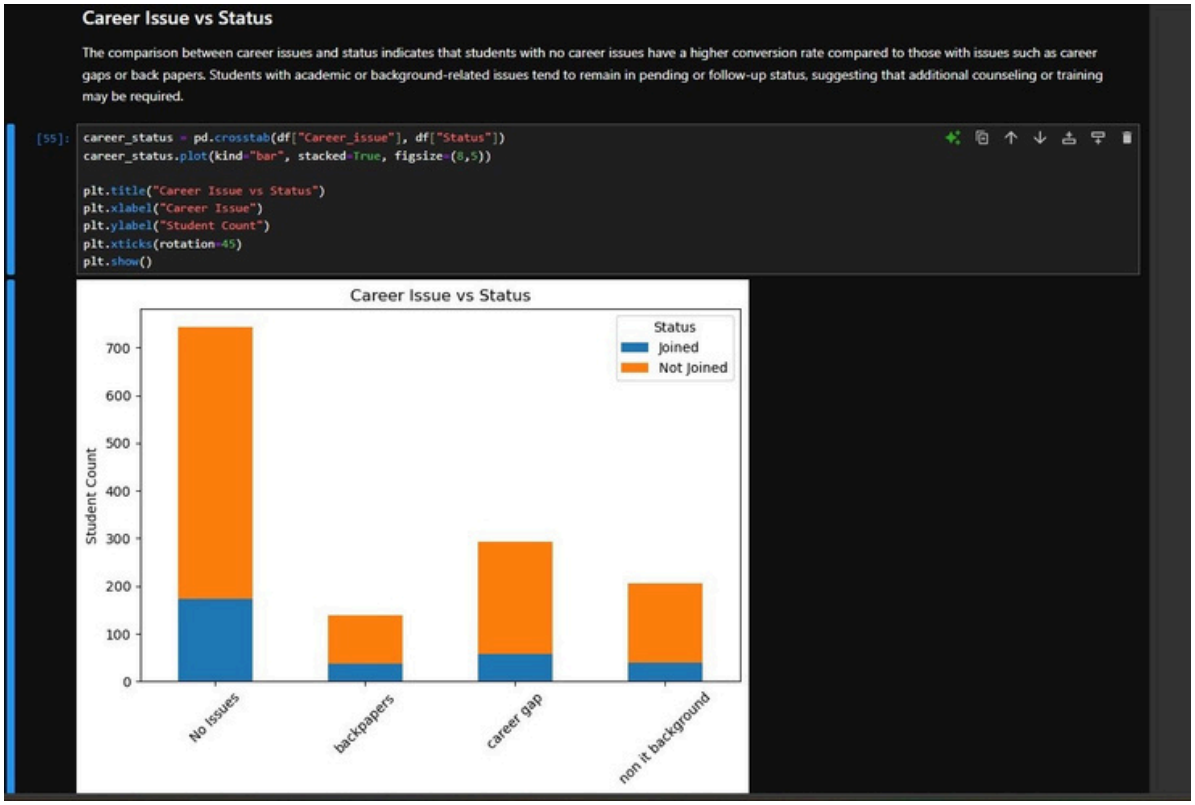


Chart 10: Career Issue vs Status — stacked (Blue = Joined, Orange = Not Joined)

Key Findings:

- 'NoIssues' students have the highest joined count (~170) but also the highest not-joined count — consistent with their larger population.
- 'Back Papers' students show very poor conversion, with only ~35 joined out of ~140 total.
- 'Career Gap' students have moderate conversion (~60 joined of ~300 total), suggesting motivation but possible financial or scheduling barriers.
- 'Non-IT Background' has a low join rate (~35 of ~205), indicating these students may need more convincing of course relevance.

SECTION 6: What Students Are Looking For

6.1 Students Looking For — Count Distribution

Thisbar chartshows thefivecategoriesstudents indicated they are looking for: others, job oriented course, short term course, IT internship, and work experience.

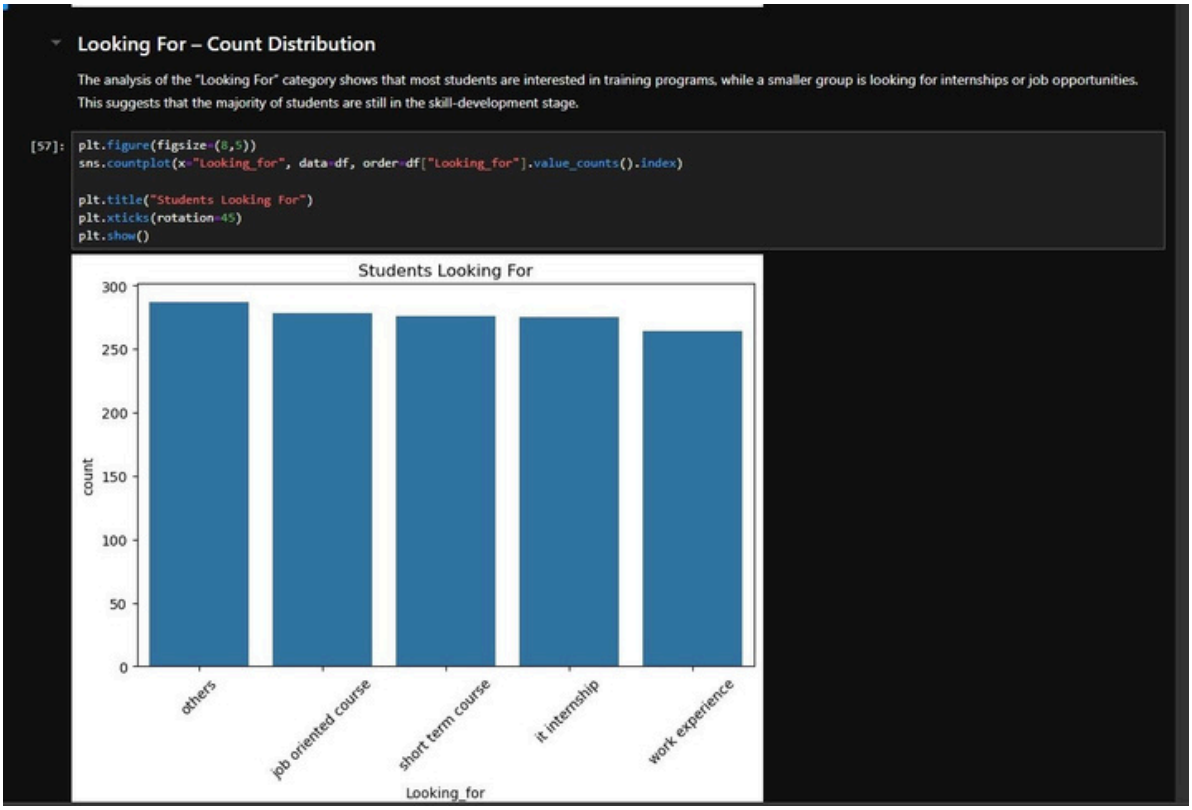


Chart 11: Students Looking For Distribution — all five categories nearly equal (~265–285 students each)

Key Findings:

- All five categories are remarkably balanced, ranging from ~265 (work experience) to ~285 (others).
- This near-equal spread suggests the institute attracts students with diverse goals and expectations.
- 'Job oriented course' and 'short term course' are strong indicators of career urgency.

6.2 Looking For vs Enrollment Status

ThisgroupedbarchartcomparesNotJoinedvsJoined students within each 'Looking For' category.

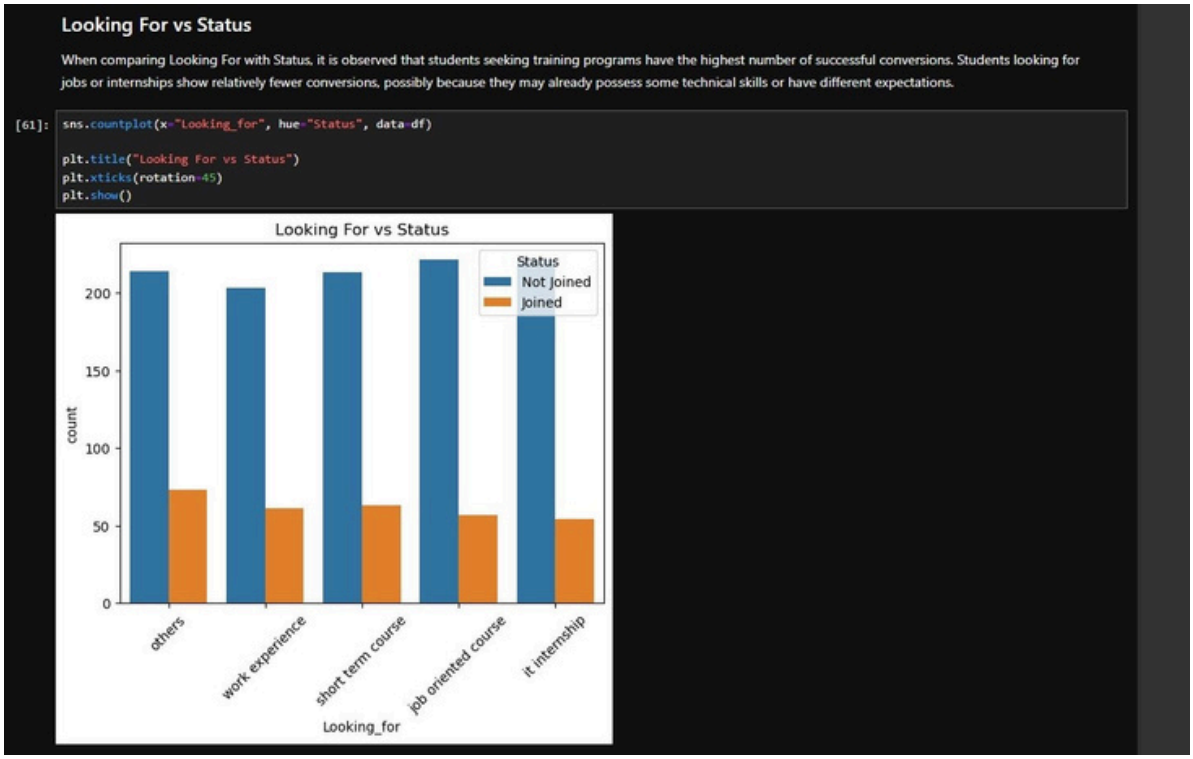


Chart 12: Looking For vs Status (Blue = Not Joined, Orange = Joined)

Key Findings:

- 'Others' category has the highest joined count (~73), slightly ahead of other categories.
- 'IT Internship' seekers show the lowest conversion (~55 joined), possibly because they are still exploring options.
- All categories show a consistently high Not Joined rate (~200+), confirming the systemic conversion challenge.

SECTION 7: Technology Preferences

7.1 Technology Distribution

Thishorizontalbarchartrankstechnologiesbystudent interest, from most to least popular.

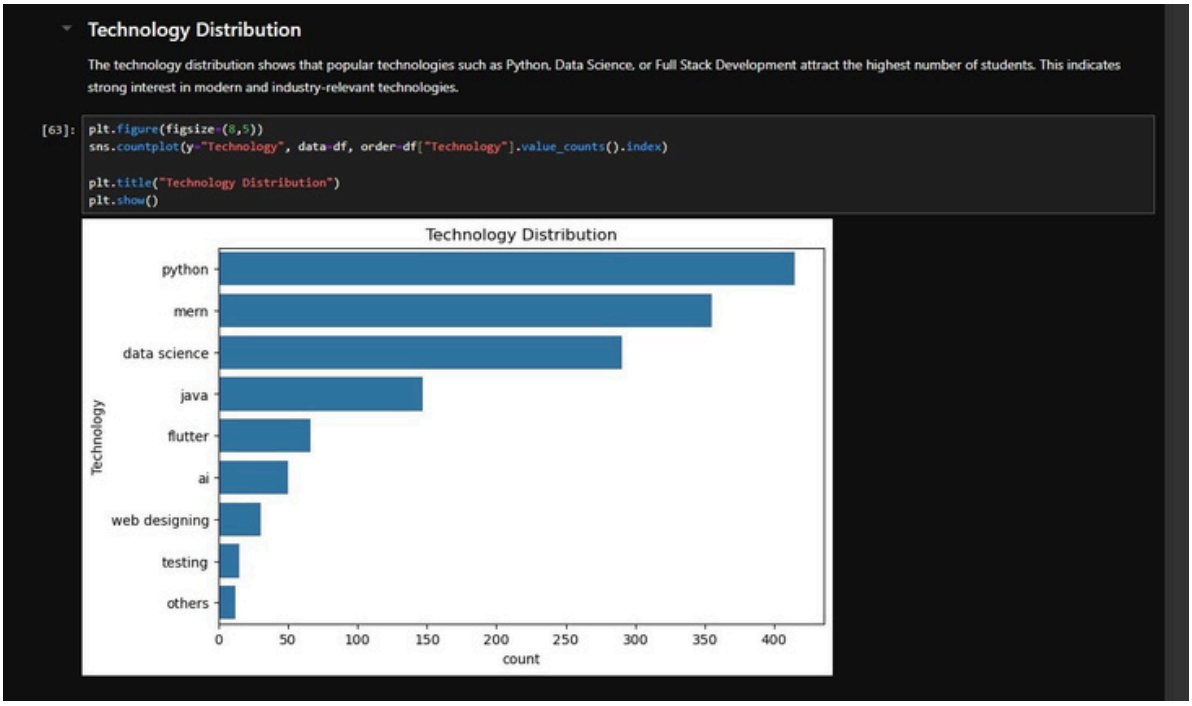


Chart 13: Technology Distribution — Python leads at ~415, followed by MERN (~350), Data Science (~295), Java (~150)

Key Findings:

- Python is the most popular technology by a significant margin (~415 students).
- MERN stack (~350) reflects strong interest in full-stack web development.
- Data Science (~295) continues to grow in demand, consistent with industry trends.
- Java (~150), Flutter (~65), AI (~55), Web Designing (~35), Testing (~20), Others (~15) follow in descending order.

7.2 Technology Selected vs Enrollment Status

Thisstackedbarchartshowshowenrollmentstatusmapsto each technology choice.

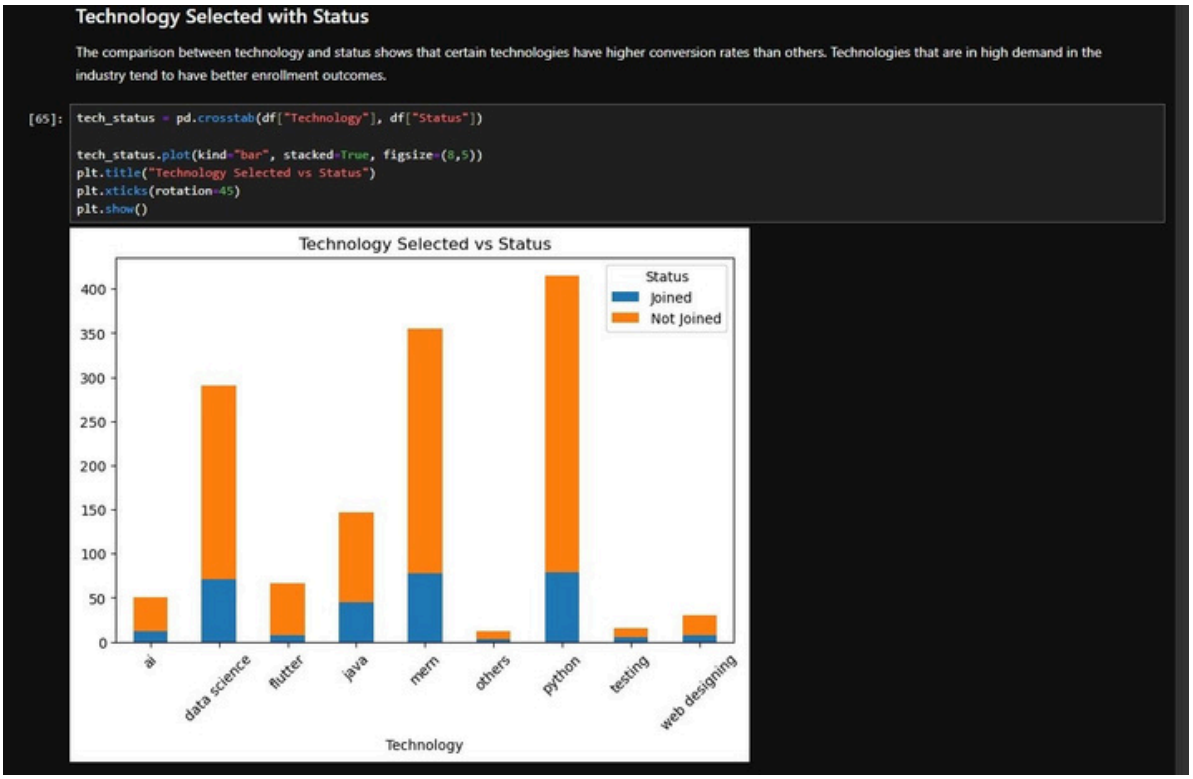


Chart 14: Technology Selected vs Status — stacked (Blue = Joined, Orange = Not Joined)

Key Findings:

- Pythonhas the highest absolute not-joined count (~340) but also strong joined numbers (~75).
- Data Science shows a proportionally better conversion rate compared to Python and MERN.
- AI, Testing, and Web Designing show very low absolute enrollments.
- Strengthening Python and MERN course quality and marketing could significantly boost overall conversions.

SECTION 8: Mode of Training & Time Slot Preferences

8.1 Mode of Training Distribution

Thedonutchartbelowshowsstudentpreferencesforhowtheywishtoreceive training.

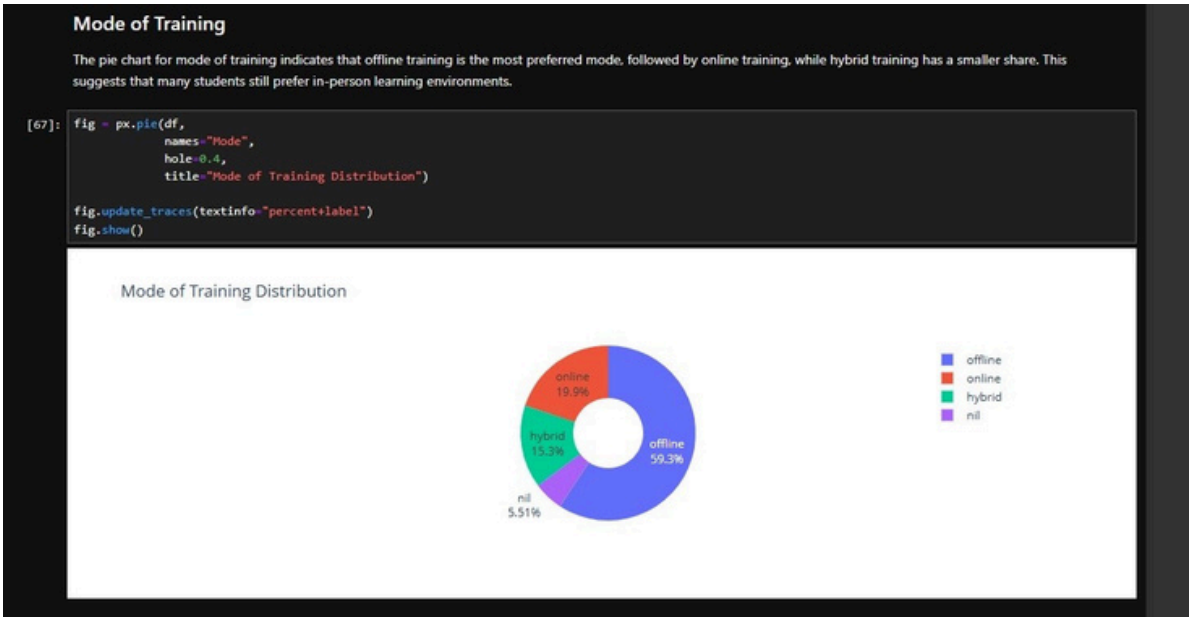


Chart 15: Mode of Training Distribution — Offline: 59.3%, Online: 19.9%, Hybrid: 15.3%, Nil: 5.51%

Key Findings:

- Offline training is overwhelmingly preferred at 59.3%, confirming demand for in-person learning.
- Online training accounts for 19.9%, reflecting a growing but secondary digital learning preference.
- Hybrid (15.3%) appeals to students who want flexibility between both modes.
- The 5.51% 'nil' response may represent undecided students requiring further counseling.

8.2 Preferred Time Slot

Thisbarchartshowswhenstudents prefer to attend training sessions.

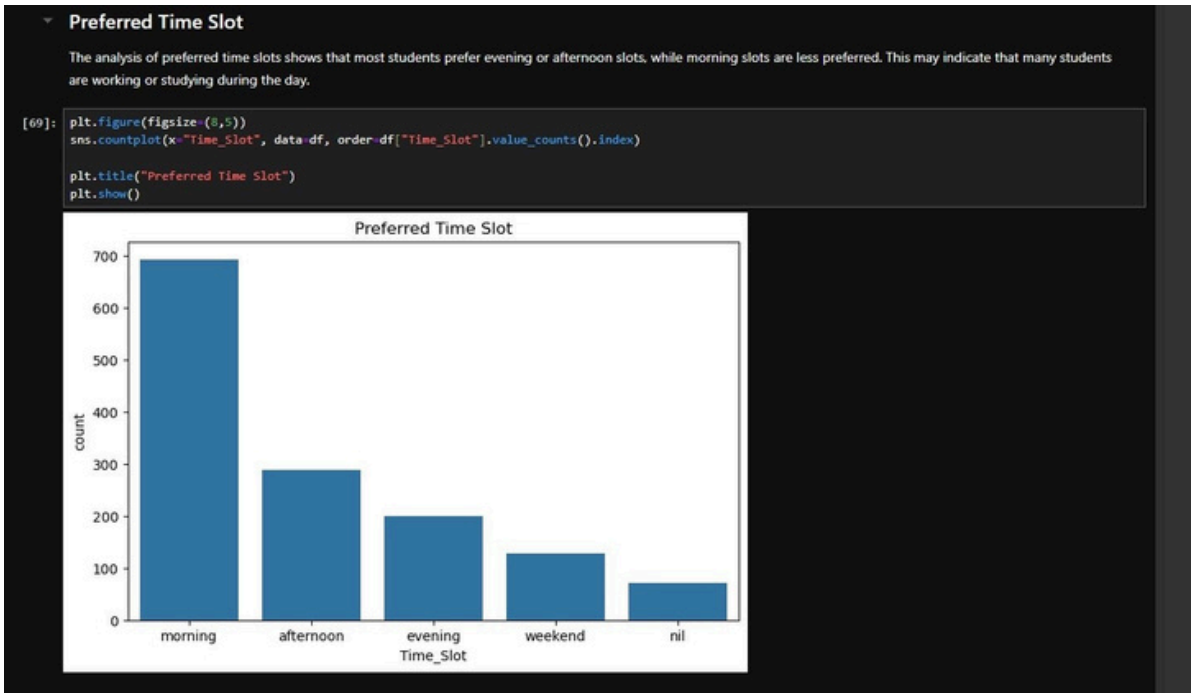


Chart 16: Preferred Time Slot — Morning: ~690, Afternoon: ~290, Evening: ~200, Weekend: ~125, Nil: ~70

Key Findings:

- Morning is the dominant time slot with ~690 students — more than double the next preference.
- Afternoon (~290) and Evening (~200) are secondary choices.
- Weekend (~125) and Nil (~70) are the least preferred slots.
- High morning preference suggests many students are between jobs or have flexible daily schedules.

8.3 Time Slot vs Status

This grouped bar chart compares joining status across each time slot preference.

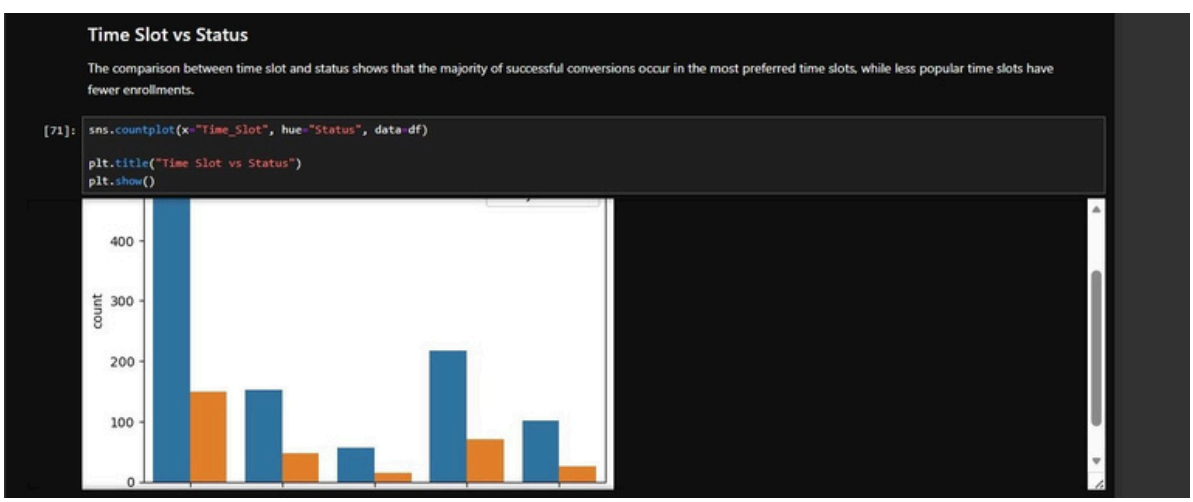


Chart 17: Time Slot vs Status (Blue = Not Joined, Orange = Joined)

Key Findings:

- Morning slots generate the highest joined count (~145), consistent with their popularity.
- Afternoon (~65) and Evening (~45) follow in conversion volume.
- Weekend slot conversions are very low (~20), suggesting lower commitment among weekend-preferring students.

8.4 Time Slot vs Status Distribution (Stacked)

This stacked bar chart confirms the morning slot's dominance in both total enquiries and conversions.

```
[73]: ts = pd.crosstab(df["Time_Slot"], df["Status"])

      ts.plot(kind="bar", stacked=True, figsize=(8,5))

      plt.title("Time Slot vs Status Distribution")
      plt.xlabel("Time Slot")
      plt.ylabel("Count")
      plt.xticks(rotation=45)
      plt.show()
```

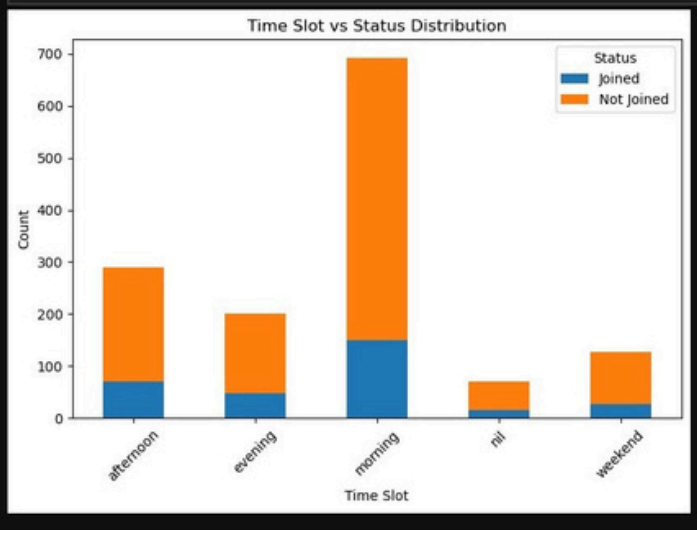


Chart 18: Time Slot vs Status Distribution — stacked (Blue = Joined, Orange = Not Joined)

Key Findings:

- Morning has the tallest bar (~690 total), with ~145 joined (blue) and ~545 not joined (orange).
- Afternoon shows ~290 total with ~65 joined — a conversion rate of ~22%.
- Evening (~200 total, ~45 joined) and Weekend (~125 total, ~20 joined) have lower absolute and proportional conversions.
- Institute should prioritize morning batch scheduling and staffing to maximize enrollment.

SECTION 9: Enquiry Sources & Staff Performance

9.1 Distribution of Enquiry Sources

Thishorizontalbarchartshowshowstudentsdiscoveredtheinstitute, ordered by enquiry volume.

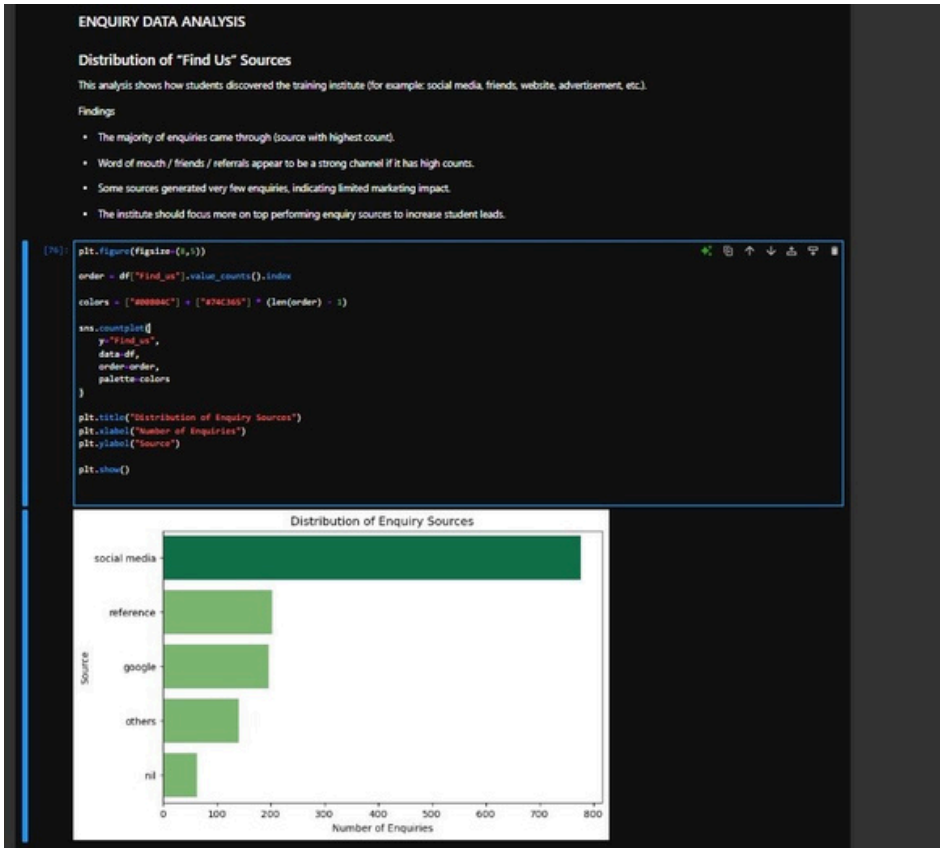


Chart 19: Distribution of Enquiry Sources — Social Media leads at ~800, Reference: ~220, Google: ~220, Others: ~180, Nil: ~80

Key Findings:

- Social Media is the primary discovery channel at ~800 enquiries — far ahead of all others.
- Reference (word of mouth) and Google each account for ~220 enquiries.
- The strong social media presence indicates effective digital marketing but raises questions about lead quality vs. quantity.

9.2 Student Joining Status by Enquiry Source

Thisstackedbarchartcross-tabulatesenquirysourcewithjoining status to reveal channel quality.

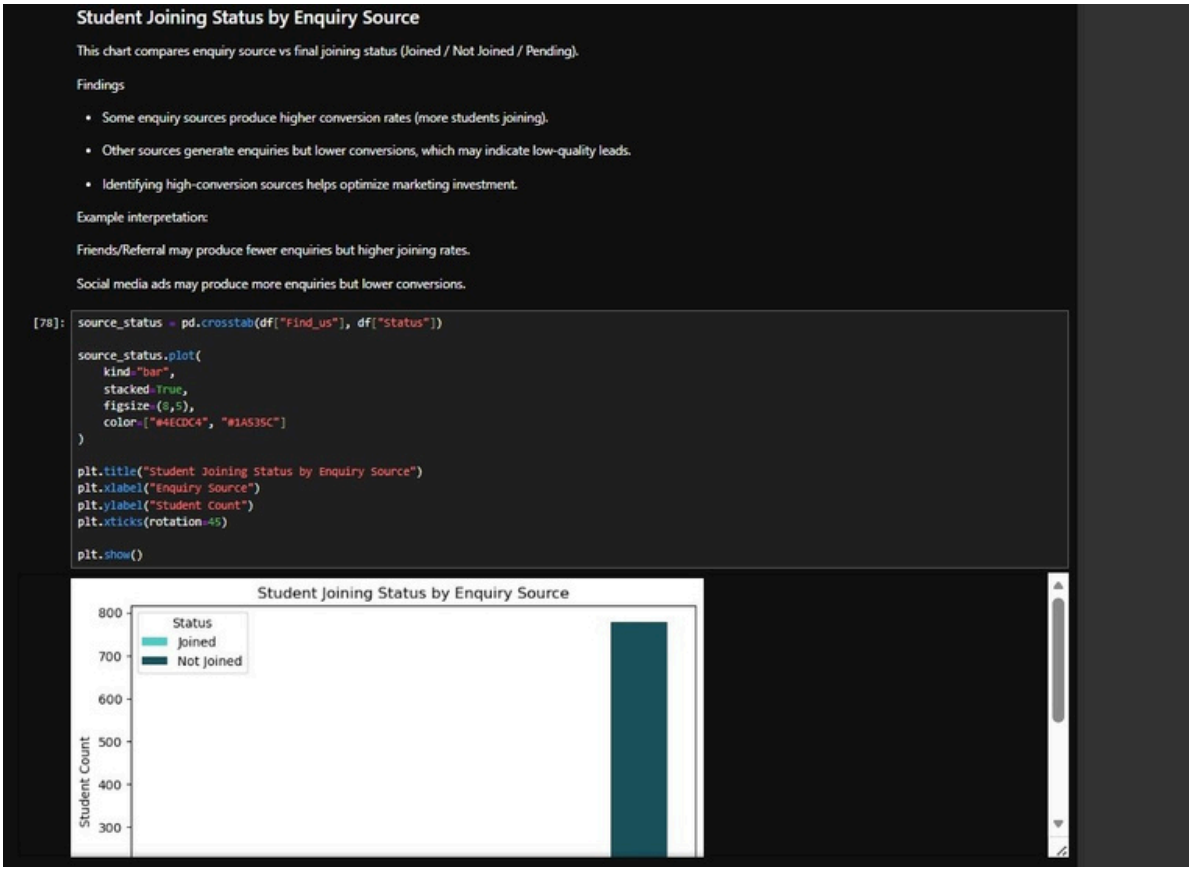


Chart 20: Student Joining Status by Enquiry Source (Light Teal = Joined, Dark Teal = Not Joined)

Key Findings:

- Social media generates the highest volume but likely has a lower conversion rate relative to its enquiry count.
- Reference-based enquiries tend to have better intent and likely yield higher conversion rates.
- Google leads represent mid-funnel searchers with moderate conversion potential.
- Investing in referral incentive programs could yield better ROI than scaling social media spend alone.

9.3 Enquiries Attended by Staff

This horizontal bar chart shows individual staff workload based on the number of enquiries handled.

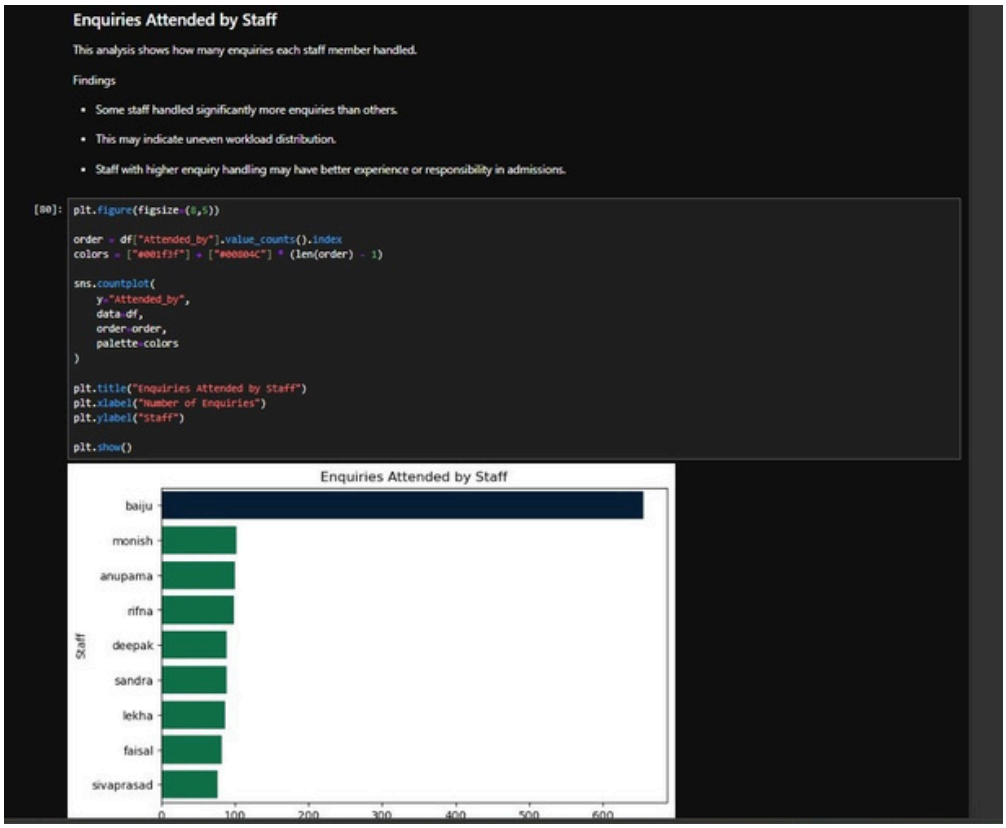


Chart 21: Enquiries Attended by Staff — Baiju handled 600+ enquiries vs. 150–220 for all other staff

Key Findings:

- Baiju handled over 600 enquiries — approximately 3–4x more than any other staff member.
- Other staff (Monish, Anupama, Rifna, Deepak, Sandra, Lekha, Faisal, Sivaprasad) each handled 150–220 enquiries.
- This extreme imbalance creates burnout risk and may impact enquiry follow-up quality.
- The institute should review enquiry routing rules and distribute workload more evenly across the team.

SECTION 10: Recommendations & Strategic Insights

10.1 Improve Conversion Rate

- Implement a 7-day structured follow-up process for all Not Joined enquiries.
- Offer free demo/trial classes to undecided students to build commitment.
- Create targeted counseling for career gap and non-IT background students addressing their specific concerns.

10.2 Optimize Technology & Course Offerings

- Double investment in Pythonand MERN courses — highest demand technologies.
- Develop dedicated Data Science tracks to capture the growing third-place demand.
- Bundle AI/ML content with Python courses to lift AI enrollment organically.

10.3 Marketing & Channel Strategy

- Launch a referral incentive program to grow the high-quality reference channel.
- Improve Google SEO/SEM to capture more intent-driven search traffic.
- Focus geographic campaigns on top 3 locations which show disproportionate interest.

10.4 Operational Improvements

- Redistribute enquiry workloadevenly — Baiju's extreme load is a significant risk.
- Increase morning batch capacity as this slot drives the highest conversions.
- Maintain and expand offline training infrastructure given the 59.3% offline preference.

10.5 Student Support Programs

- Design re-skilling programs specifically for 2017–2020 graduates seeking career transitions.
- Create weekend-specific batch offerings to serve that underserved segment.
- Develop flexible payment or scholarship schemes to improve conversion among back-paper and career-gap students.

--- End of Report ---